

Vertebra FEA Quickstart

Ogo spine compression workflow on the ogo-fea branch

This guide shows how to check out the Ogo FEA branch and run a solved vertebra compression model with FAIM/N88. It assumes a density-calibrated CT image, a labelled vertebra segmentation, and access to the Numerics88 command-line tools.

Workflow Summary

- Extract vertebral body and posterior process labels from the spine mask.
- Crop, ICP-align to the bundled Ogo vertebral reference, and resample to isotropic spacing.
- Build superior and inferior PMMA supports on stable body contact surfaces.
- Convert calibrated density to material IDs, write an .n88model, audit boundary conditions, solve with FAIM/N88, and write a compact results CSV.
- Optionally transform an ROI mask into model space and report masked Pistoia in addition to full-model Pistoia.

Checkout And Install

```
git clone git@github.com:Bonelab/Ogo.git
cd Ogo
git checkout ogo-fea
git pull origin ogo-fea

conda activate ogo
python -m pip install -e .
ogoFEA --help
```

Required Inputs

Input	Requirement
Calibrated CT	Scalar density image in K2HPO4-equivalent units.
Labelled spine mask	Integer labels containing the vertebral body and posterior process.
Optional ROI mask	Binary mask in the original CT image space, for example an L1 body mask for masked Pistoia.

The vertebra target is passed as LEVEL:BODY_LABEL:PROCESS_LABEL. Example: L1:2:3 means output level L1, body label 2, and posterior process label 3.

Standard L1 Command

```
ogoFEA spine \
  sub-001_desc-vqct_ct.nii.gz \
  sub-001_desc-spine_labels.nii.gz \
  --vertebra L1:2:3 \
  --output_path derivatives/fea \
  --threads 4 \
  --faim_bin_dir /path/to/n88/bin \
  --run_pistoia \
  --critical_volume 12 \
  --critical_strain 0.007
```

Use --threads 4 for a typical single-model run. This is the FAIM solver thread count for that one model; it does not launch multiple subjects in parallel.

Full And Masked Pistoia

```
ogoFEA spine \
  sub-001_desc-vqct_ct.nii.gz \
  sub-001_desc-spine_labels.nii.gz \
  --vertebra L1:2:3 \
  --pistoia_mask sub-001_desc-L1Body_mask.nii.gz \
  --output_path derivatives/fea \
  --threads 4 \
  --faim_bin_dir /path/to/n88/bin \
  --run_pistoia \
  --critical_volume 12 \
  --critical_strain 0.007
```

Result	Description
Full-model Pistoia	Uses the solved vertebra model and excludes PMMA material.
Masked Pistoia	Uses only elements selected by the supplied ROI mask after transformation into model space.

In one ogoFEA run, the same critical volume and critical strain are used for full-model and masked Pistoia. The results CSV includes `masked_pistoia_*` fields when a mask is supplied.

N88 And FAIM Path Handling

Ogo does not search the entire filesystem for N88. Commands are resolved in this order:

- Explicit command overrides such as `--faim_command` or `--n88pistoia_command`.
- `--faim_bin_dir`, for example `/path/to/n88/bin`.
- `--faim_install_root`, where Ogo checks common subdirectories.
- The current shell `PATH`.

```
ogoFEA spine image.nii.gz labels.nii.gz \
  --vertebra L1:2:3 \
  --output_path derivatives/fea \
  --faim_env ogoloco-n88 \
  --threads 4
```

Expected Outputs

File	Meaning
*_L1.n88model	Generated finite-element model.
*_L1_modeling.json	Traceable metadata for inputs, preprocessing, materials, boundary conditions, and reporting.
*_L1_results.csv	Compact one-row results table for downstream analysis.
*_L1_pistoia.txt	Full-model Pistoia report, when Pistoia is run.
*_L1_masked_pistoia.txt	ROI-masked Pistoia report, when <code>--pistoia_mask</code> is supplied.
*_L1_pistoia_mask.nii.gz	ROI mask transformed into model space.

Model Defaults

Setting	Default
Model type	Linear elastic vertebral compression.
High-level preset	benchmark-linear.
Resampling	1.0 mm isotropic.
Density-to-modulus law	$E = 2980 * \rho^{1.05}$ MPa for benchmark-linear spine models.
Bone Poisson ratio	0.3.
PMMA material	$E = 2500$ MPa, Poisson ratio = 0.3.
PMMA geometry	10 mm thickness with stable-contact support placement.
Default endpoint	0.68% compression strain, converted to millimeters before solving.

Quick Quality Checks

```
ogoFEA spine \  
  sub-001_desc-vqct_ct.nii.gz \  
  sub-001_desc-spine_labels.nii.gz \  
  --vertebra L1:2:3 \  
  --output_path derivatives/fea \  
  --debug \  
  --no-solve
```

- The vertebral body should be upright in the model frame.
- The posterior process should be posterior to the body, not rotated into the axial direction.
- Superior and inferior PMMA supports should contact stable body surfaces rather than isolated osteophytes.
- The boundary-condition audit should pass.
- The modeling JSON should record the expected labels, material law, endpoint, and N88 settings.

Common Problems

Symptom	Likely cause	What to check
Cannot find faim or n88pistoia	N88 tools are not on PATH.	Pass <code>--faim_bin_dir</code> , <code>--faim_env</code> , or explicit command paths.
Missing masked Pistoia fields	No ROI mask was supplied, or masked Pistoia failed.	Check <code>*_results.csv</code> , <code>*_masked_pistoia.txt</code> , and the log.
Process appears rotated	ICP alignment or label definition problem.	Run <code>--debug --no-solve</code> and inspect the model.
PMMA contacts osteophytes	Endplate surface is irregular.	Inspect debug output and modeling JSON; exclude cases that fail QC.
Existing solution is reused	The <code>.n88model</code> already contains a solution.	Use a clean output directory before rerunning.

Minimal Reproducible Handoff

```
git clone git@github.com:Bonelab/Ogo.git
cd Ogo
git checkout ogo-fea
python -m pip install -e .
```

```
ogoFEA spine \
  image_K2HP04.nii.gz \
  spine_labels.nii.gz \
  --vertebra L1:2:3 \
  --pistoia_mask L1_body_mask.nii.gz \
  --output_path derivatives/fea \
  --threads 4 \
  --faim_bin_dir /path/to/n88/bin \
  --run_pistoia \
  --critical_volume 12 \
  --critical_strain 0.007
```

For tabulated analysis, use derivatives/fea/*_L1_results.csv. For model inspection, use derivatives/fea/*_L1.n88model.